

Muhammad Ali Waris Khan

AI & Backend Engineer | LLM & RAG Developer

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🐙 github.com/maw-khan 🔗 LinkedIn 🌐 Portfolio

Professional Summary

AI & Backend Engineer specializing in Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and intelligent document processing systems. Experienced in designing and deploying production-style AI applications using Python, LangChain, Gemini AI, FAISS, REST APIs, and modular backend architectures. Skilled in document ingestion, semantic retrieval, vector search, conversational AI, and scalable backend development. Hands-on experience building end-to-end AI systems through self-driven projects and academic work, with working knowledge of Azure AI architecture and cloud-native document intelligence solutions.

Technical Skills

- **Programming:** Python, SQL, AsyncIO (Async/Await), Threading, Concurrent Programming
- **Backend Engineering:** Python Services, REST APIs, HTTP, JSON, PostgreSQL, Modular Architecture, Exception Handling
- **LLM Engineering:** Gemini AI, LangChain, Prompt Engineering, Conversational AI, RAG Pipelines, Hallucination Mitigation
- **Document Intelligence:** PDF Parsing (PyMuPDF), Chunking, Enrichment, Vectorization, Indexing
- **Retrieval & Vector Search:** FAISS, Semantic Search, Dense Embeddings, MMR Retrieval, Cross-Encoder Reranking
- **Machine Learning / NLP:** Sentence Transformers, Hugging Face Embeddings, NLP Preprocessing
- **Frontend & Deployment:** Streamlit, Streamlit Cloud; Familiar with Azure cloud concepts including Blob Storage, Azure AI Search, Function Apps, and Azure OpenAI architecture.
- **Software Engineering:** Debugging, Stateful Session Management, Logging, Reproducible Pipelines, Production-Oriented Development
- **Development Tools:** Git, GitHub, VS Code, Cursor AI, Postman
- **Libraries & Tools:** PyMuPDF, yt-dlp, youtube-transcript-api, sentence-transformers, dotenv

Professional Experience

Lab Engineer (Academic)
(Academic Appointment)

NUST, Islamabad

- Assisted faculty members during laboratory sessions and practical coursework.
- Guided students in programming, debugging, and technical problem-solving.
- Supported laboratory operations, software setup, and computing resources.
- Collaborated with faculty to facilitate practical learning and technical demonstrations.

Deployed Projects

1. Production-Style Document Intelligence Platform

(Python, LangChain, Gemini AI, FAISS, Streamlit, PyMuPDF)

- Engineered and deployed a production-style document intelligence platform implementing document extraction, enrichment, chunking, embedding generation, vectorization, indexing, and semantic retrieval for multi-PDF interaction.
- Built a multi-stage retrieval pipeline using Gemini embeddings, FAISS, MMR retrieval, and Cross-Encoder reranking to improve search relevance and grounded responses.
- Developed modular Python services for document ingestion, indexing, retrieval, conversational memory, and response generation following reusable backend architecture principles.
- Implemented streaming responses, page-level citations, source traceability, and conversational memory for explainable multi-turn question answering.
- Improved pipeline reliability through validation, exception handling, and logging across document ingestion and retrieval workflows.
- Deployed the application on Streamlit Cloud. Live Demo: pdfcantalk.streamlit.app
- GitHub: github.com/maw-khan/context-aware-rag-chatbot-gemini

2. AI Resume Analyzer with ATS Scoring

(Python, Gemini AI, Streamlit, PyMuPDF, Regex NLP)

- Developed AI-powered ATS resume analyzer evaluating resumes against job descriptions through NLP keyword analysis and skill matching
- Built robust automated PDF parsing and preprocessing pipeline with validation and exception handling
- Engineered custom ATS scoring algorithm with missing-skill detection and compatibility scoring
- Integrated Gemini AI for contextual optimization feedback and recruiter-style recommendations
- Implemented explainable interface displaying matched/missing keywords and improvement suggestions
- Designed modular backend architecture with reusable Python components and secure API integration.
- Deployed on Streamlit Cloud. Live Demo: atsresume-analyzer.streamlit.app
- GitHub: github.com/maw-khan/resume-analyzer-ai

3. Transcript Intelligence RAG Platform

(Python, Gemini AI, LangChain, FAISS, Sentence Transformers, Streamlit)

- Engineered a transcript intelligence platform for semantic interaction with YouTube video transcripts.
- Implemented transcript intelligence pipeline with metadata extraction, timestamp-aware chunking, and semantic indexing
- Engineered multi-stage retrieval combining FAISS, MMR, and Cross-Encoder reranking for high precision
- Integrated conversational memory and streaming LLM responses for real-time multi-turn interaction
- Developed timestamp-based citation system linking answers directly to original video segments
- Implemented persistent vector storage with incremental indexing for scalable workflows
- Deployed on Streamlit Cloud. Live Demo: chatwithvideos.streamlit.app
- GitHub: github.com/maw-khan/youtube-rag-chatbot

4. Academic Document Intelligence Platform

(Python, Gemini AI, LangChain, FAISS, Hugging Face, Streamlit)

- Developed an academic document intelligence platform for semantic search, literature review generation, and research paper analysis.
- Built multi-document RAG pipeline with Hugging Face embeddings and FAISS for semantic retrieval

- Implemented multi-stage retrieval with reranking for improved precision and grounded responses
- Engineered automated summarization, methodology extraction, and literature synthesis using Gemini AI
- Applied hallucination-aware prompt engineering for context-grounded academic question answering
- Built modular production architecture with explainable citations and robust validation
- Deployed on Streamlit Cloud. Live Demo: chatwithresearchpaper.streamlit.app
- GitHub: <https://github.com/maw-khan/research-paper-assistant>

Education

Masters in Artificial Intelligence

College of Electrical and Mechanical Engineering, NUST, Islamabad